

Formalizing the homotopy extension property in LEAN

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Zusammenfassung

Motivation: Whitehead theorem oder so?

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1 Vorbemerkung

In this work, if not specified in any other way, every space has a topology and every map is a continuous map.

1.1 Motivation

The goal of this thesis is to formalize the *homotopy extension property (HEP)* in Lean. I started with the proof of the *retraction criterion* and worked towards the result that every relative CW complex satisfies the HEP. Our motivation for studying this property is that it is needed for two major theorems in algebraic topology. The first is the *cellular approximation theorem*.

Theorem 1.1 (Cellular Approximation Theorem, [MR1867354]). *Every map $f: X \rightarrow Y$ of CW complexes is homotopic to a cellular map. If f is already cellular on a subcomplex $A \subset X$, the homotopy may be taken to be stationary on A .*

Here a map is called *cellular* if it sends the n -skeleton of X to the n -skeleton of Y for all $n \in \mathbb{N}$.

Equipping S^k with the CW structure consisting of one 0-cell and one k -cell, it has no cells in dimensions $0 < n < k$. Hence any map $S^n \rightarrow S^k$ with $n < k$ is homotopic to one landing in the n -skeleton, which is a single point, and is therefore nullhomotopic. This yields the corollary $\pi_n(S^k) = 0$ for all $n < k$.

Secondly, the result that relative CW complexes satisfy the HEP is one of the main steps towards the proof of *Whitehead's theorem*.

Theorem 1.2 (Whitehead's Theorem, [MR1867354]). *If a map $f: X \rightarrow Y$ between connected CW complexes induces isomorphisms $f_*: \pi_n(X) \rightarrow \pi_n(Y)$ for all n , then f is a homotopy equivalence. In case f is the inclusion of a subcomplex $X \hookrightarrow Y$, the conclusion is stronger: X is a deformation retract of Y .*

Whitehead's theorem states that, in order to prove that two connected CW complexes are homotopy equivalent, it suffices to exhibit a map between them that induces isomorphisms on all homotopy groups, i.e. a weak homotopy equivalence. Showing that two spaces are homotopy equivalent is in general a difficult task, and Whitehead's theorem provides a powerful tool for such problems.

2 Homotopy Extension Property

2.1 The Homotopy Extension Property

There are several equivalent ways to define the homotopy extension property, which will be discussed in this first chapter. This first definition is closest to what I formalized in Lean.

Definition 2.1. A pair of topological spaces (X, A) with $A \subseteq X$ has the *homotopy extension property (HEP)* if for every topological space Y , every map $f: X \rightarrow Y$, and every homotopy $H: A \times [0, 1] \rightarrow Y$ satisfying $H(a, 0) = f(a)$ for all $a \in A$, there exists a homotopy $H': X \times [0, 1] \rightarrow Y$ such that

$$\begin{aligned} H'(x, 0) &= f(x) && \text{for all } x \in X, \\ H'(a, t) &= H(a, t) && \text{for all } (a, t) \in A \times [0, 1]. \end{aligned}$$

Remark 2.2. Note, that for any space X , the pair (X, X) and the pair $(X, \emptyset)$ have the *HEP*.

– unsauber, gute Quelle finden

Definition 2.3. A pair of two maps (f, H) , which agree on the space A respectively $A \times \{0\}$, same as in the previous definition, is an *extension problem*.

Remark 2.4. Another equivalent way of phrasing this definition is to say that (X, A) has the *HEP* if and only if every map defined on $X \cup_A (A \times [0, 1])$ extends to a map on $X \times [0, 1]$. Note that this is, in general, not the same as requiring that every map defined on $X \times \{0\} \cup (A \times [0, 1])$ extends to a map on $X \times [0, 1]$. This is because the natural map

$$X \cup_A (A \times [0, 1]) \rightarrow X \times \{0\} \cup (A \times [0, 1])$$

is not necessarily a homeomorphism. In general, it is only a continuous bijection. However, if A is closed, then this map is a homeomorphism and therefore yields another equivalent definition.

The equivalence from theorem 2.4 is proven in the following lemma.

Lemma 2.5. *Let A be a subset of a topological space X .*

(i) (X, A) has the HEP

(ii) *for every topological space Y and every continuous map $g: X \times \{0\} \cup A \times [0, 1] \rightarrow Y$ there exists an extension of g to a map $G: X \times [0, 1] \rightarrow Y$*

Then (i) $\implies$ (ii) and under the additional condition, that A is closed, also the inverse direction (ii) $\implies$ (i) holds.

Proof. (i) $\Rightarrow$ (ii): Let $g : X \times \{0\} \cup A \times [0, 1] \rightarrow Y$. We define maps $f : X \rightarrow Y, x \mapsto g(x, 0)$ and $H : A \times [0, 1] \rightarrow Y, (x, t) \mapsto g(x, t)$. It is easy to check, that f and H agree on $A \times [0, 1]$, so the *HEP* for the pair (X, A) yields an extension of those two maps. We can chose G to be exactly this extended homotopy.

(A closed $\wedge$ (ii)) $\Rightarrow$ (i): Let $f : X \rightarrow Y$ and $H : A \times [0, 1] \rightarrow Y$. We define the map

$$g : X \times \{0\} \cup A \times [0, 1] \rightarrow Y, (x, t) \mapsto \begin{cases} f(x), & \text{if } t = 0, \\ H(x, t), & \text{else} \end{cases}$$

Clearly g is well-defined, since f and H agree on A . To proof continuity of g , we need that $A \subseteq X$ is closed:

$A \subseteq X$ closed $\implies A \times [0, 1] \subset X \times [0, 1]$ closed. Furthermore, $X \times \{0\} \subseteq X \times [0, 1]$ is closed. So g is continuous, because it is continuous on two closed subsets and therefor also on their union.

By assumption there exists an extension $G : X \times [0, 1]$ of g , which we can choose to be the extended homotopy for the definition of the *HEP*. $\square$

Lemma 2.6. *Let A be a subset of a topological space X . Then the following are equivalent:*

(i) *for every $g : A \rightarrow Y$ there exists an extension $G : X \rightarrow Y$*

(ii) *A is a retract of X , i.e. $\exists r : X \rightarrow A$, such that r is the identity on $A \subset X$.*

Proof. (i) $\Rightarrow$ (ii): We look at the special case of (i), when $Y = A$ and $g = id_A$. Then the extension G is the desired retraction.

(ii) $\Rightarrow$ (i): Let $g : A \rightarrow Y$. Then we define $G : X \rightarrow Y$ to be the map g precomposed with a retraction $r : X \rightarrow A$. Since r is the identity on A this indeed is an extension of g . $\square$

To proof, that a space pair actually satisfies the *HEP* it is a convenient way to use the *retraction criterion*. This follows from the previous two lemma. Often, the retraction criterion is only phrased for closed subsets $A \subset X$, but I assumed the closeness condition on A , as in theorem 2.5, only for the implication, where it is needed.

Definition 2.7. Let X be a topological space and $A \subset X$ a subset. A retraction is a continuous map $r : X \rightarrow A$, such that precomposition with the inclusion $i : A \hookrightarrow X$ yields the identity on A : $r \circ i : A \rightarrow A = id_A$.

Lemma 2.8 (Retraction Criterion). *Let X be a topological space and $A \subseteq X$ a subspace. Consider the following two statements:*

(i) *the pair (X, A) has the *HEP**

(ii) the subspace $X \times \{0\} \cup A \times [0, 1] \subseteq X \times [0, 1]$ is a retract of $X \times [0, 1]$

Then (1) $\implies$ (2). If, in addition, A is closed in X , the converse (2) $\implies$ (1) holds as well.

Proof. (i) $\implies$ (ii): If (X, A) has the HEP, then by "(i) $\implies$ (ii)" from theorem 2.5 we get: for every map $g: X \times \{0\} \cup A \times [0, 1] \rightarrow Y$ there exists an extension of g to a map on $X \times [0, 1]$. Now the statement follows directly from theorem 2.5.

(A closed $\wedge$ (ii)) $\implies$ (i): For this direction we first use that $X \cup A \times [0, 1] \subseteq X \times [0, 1]$ is closed and apply the backward direction of theorem 2.6. This yields the same intermediate step as in the previous part of this proof / the forward direction. With the same closedness as before, we apply the backward direction of theorem 2.5 and conclude the proof. $\square$

It follows from the retraction criterion, that the HEP is preserved under taking the product with a space.

Corollary 2.9. *If (X, A) has the HEP, so does $(W \times X, W \times A)$ for any space W .*

Proof. If $r: X \times [0, 1] \rightarrow X \times \{0\} \cup A \times [0, 1] \subseteq X \times [0, 1]$ is a retract, so is

$id_W \times r: W \times X \times [0, 1] \rightarrow W \times X \times \{0\} \cup W \times A \times [0, 1] \subseteq W \times X \times [0, 1]$.

Continuity follows, because id_W is the product of two continuous maps. Furthermore id_W is fixed, where required, since id_W is fixed on all of W . $\square$

-> ToDo: noch in Lean schreiben

Corollary 2.10. *If (X, A) has the HEP and J is a discret set, then $(J \times X, J \times A)$ has the HEP.*

Proof. Let $r: X \times [0, 1] \rightarrow X \times \{0\} \cup A \times [0, 1] \subseteq X \times [0, 1]$ be a retract, then also

$$\begin{aligned} id_J \times r: J \times X \times [0, 1] &\rightarrow J \times (X \times \{0\} \cup A \times [0, 1]) \\ &= J \times X \times \{0\} \cup J \times A \times [0, 1] \end{aligned}$$

is a retract. $\square$

Theorem 2.11. *The HEP is preserved under homeomorphisms, i.e. given two space pairs (X, A) and (Y, B) . If there exists a homeomorphism from X to Y , that maps A to B , and (X, A) has the HEP, then so does (Y, B) .*

Proof. Let $\phi: X \rightarrow Y$, be a homeomorphism, that maps A to B and let $r_X: X \times [0, 1] \rightarrow X \times \{0\} \cup A \times [0, 1]$ be a retraction, which we obtain from the Retraction Criterion for the pair (X, A) . $pr_X: X \times [0, 1] \rightarrow X$ and

$pr_I: X \times [0, 1] \rightarrow X$ are the projections onto the first respectively second component. We use those maps, to define r_Y .

$$\begin{aligned} r_Y: Y \times [0, 1] &\rightarrow Y \times \{0\} \cup B \times [0, 1] \\ (y, t) &\mapsto (\phi \circ pr_X \circ r_X(\phi^{-1}(y), t), pr_I \circ r_X(\phi^{-1}(y), t)) \end{aligned}$$

It is left to check, that r_Y is a retraction, which yields the *HEP* for the space pair (Y, B) . The map r_Y ...

- is continuous: the two component functions are compositions of continuous maps.
- is fixed on $Y \times \{0\} \cup B \times [0, 1]$: Let (y, t) be a point in this subspace of $Y \times [0, 1]$. Then, $(\phi^{-1}(y), t)$ is in $X \times \{0\} \cup A \times [0, 1]$, because ϕ^{-1} maps Y to X and B to A . So $(\phi^{-1}(y), t)$ lies in exactly the space, which is fixed under r_X . After applying ϕ onto the first component, we observe, that $r_Y(y, t) = (y, t)$.
- has image in $Y \times \{0\} \cup B \times [0, 1]$: Let (y, t) be a point in $Y \times [0, 1]$. $r_X(\phi^{-1}(y), t)$ is in the image of r_X , which is $X \times \{0\} \cup A \times [0, 1]$. It follows from the definition of r_Y , that either the second component of $r_Y(y, t)$ is zero, or the first one lies in B .

□

2.2 HEP for relative CW-complexes

To proof the *HEP* for relative CW-complexes, we will generalize the space pair in three steps and proof the *HEP* for each of these pairs. The general strategy is as follows.

1. Any disc relative its boundary has the *HEP*.
2. The pair of two consecutive skeletons has the *HEP*, i.e (X_{n+1}, X_n) has the *HEP* for all n .
3. Any relative CW-complex has the *HEP*.

Before we start proofing the *HEP* for any space pair, I will give some definitinons, introduce notation for CW-complexes and proof some basic properties for them, which are not yet in mathlib.

I copied the following definition of CW-complexes from the bachelor thesis of Hannah Scholz [Sch24]. Her formalisation of CW complexes, which can be found as the "classical CW-complex" in mathlib, is explained in this thesis. She chose the historical definition of CW-complexes, presented by Whithead in [Whi18].

Definition 2.12. Let X be a Hausdorff space. A *CW-complex* on X consists of a family of indexing sets $(I_n)_{n \in \mathbb{N}}$ and a family of maps $(Q_i^n : D_i^n \rightarrow X)_{n \geq 0, i \in I_n}$ s.t.

1. $Q_i^n|_{D_i^n} : D_i^n \rightarrow Q_i^n(D_i^n)$ is a homeomorphism. We call $e_i^n := Q_i^n(D_i^n)$ an *(open) n -cell* (or a cell of dimension n) and $\bar{e}_i^n := Q_i^n(D_i^n)$ a *closed n -cell*.
2. For all $n, m \in \mathbb{N}$, $i \in I_n$ and $j \in I_m$ where $(n, i) \neq (m, j)$ the cells e_i^n and e_j^m are disjoint.
3. For each $n \in \mathbb{N}$, $i \in I_n$, $Q_i^n(\partial D_i^n)$ is contained in the union of a finite number of closed cells of dimension less than n .
4. $A \subseteq X$ is closed iff $Q_i^n(D_i^n) \cap A$ is closed for all $n \in \mathbb{N}$ and $i \in I_n$.
5. $\bigcup_{n \geq 0} \bigcup_{i \in I_n} Q_i^n(D_i^n) = X$.

We call Q_i^n a *characteristic map* and $\partial e_i^n := Q_i^n(\partial D_i^n)$ the *frontier of the n -cell* for any i and n . Additionally we define $X_n := \bigcup_{m < n+1} \bigcup_{i \in I_m} \bar{e}_i^m$ and call it the *n -skeleton* of X for $-1 \leq n \leq \infty$

Definition 2.13. CW-complex Hanna – ToDo

– Charakteristic map

The more modern way to think about CW-complexes is the following: – ToDo

The modern definition yields quotientmaps from every D^n to a corresponding cell of dimension n . The weak topology on a CW complex agrees with the quotient-topology induced by those quotientmaps. – ToDo : Könnte man auch als Theorem mit reinnehmen, aber brauche ich im Finalen Beiweis vermutlich nicht – Furthermore, we can define a quotient map onto all of the n -skeleton of a relative CW complex in the following way : is a quotientspace of the disjoint union of the base space together with all D^k in dimension $k \leq n$. We will proof, that the weak topology of the relative CW complex indeed is the same as the topology induced by this quotientmap onto the skeleton.

Lemma 2.14. Let (X, C) be a relative CW-complex with skeleton X_n . There exists a quotient map $C \dot{\cup} \coprod_{n \leq m} \coprod_{i \in J_n} D^n \rightarrow X_m$ for every $m \in \mathbb{N}$.

Proof. We define the map

$$Q : C \dot{\cup} \coprod_{n \leq m} \coprod_{i \in J_n} D^n \rightarrow X_m$$

$$x \mapsto \begin{cases} Q_i^n(x), & \text{if } x \in \text{the } i\text{th copy of } D^n, \\ x, & \text{else } (x \in C) \end{cases}$$

Surjectivity of Q follows directly from the modern definition, and it is also already a theorem in mathlib, which writes the m -skeleton as the union of the base-space C together with all cells in dimension up to dimension m . To prove, Q defines a quotient map, it remains to prove that the topologies agree.

For $A \subset X_m$ closed in the weak topology of a relative CW complex, we get by definition, that $Q_i^n(D^n) \cap C$ is closed for all $n \in \mathbb{N}$ and $i \in J_n$ and $A \cap C$ closed. Together with the characteristic maps and the identity being continuous, this is all we need, for $Q^{-1}(A)$ to be closed as well.

Assume $A \subset X_m$ closed in the topology, which is induced by Q . □

Theorem 2.15. $\forall n \in \mathbb{N}$: $(D^m, \partial D^m)$ has the HEP.

There are two different attempts to prove this theorem. One is quite visual via the retraction criterion and the other one constructs the extended homotopy explicitly. Here, I will give the constructive proof, because that one was easier to formalize in LEAN.

Proof. Given a homotopy $H : \partial D^m \times [0, 1] \rightarrow Y$ and a map $f : D^m \rightarrow Y$, which agree on $\partial D^m \times \{0\}$ identified with ∂D^m . We assemble these to a map $h : D_2^m \rightarrow Y$, where D_2^m denotes a disc with radius 2 in $\mathbb{R}^m$. h is defined in the following way.

$$h(x) = \begin{cases} f(x) & \text{if } \|x\| \leq 1 \\ H(x/\|x\|, \|x\| - 1) & \text{else} \end{cases}$$

Using h we define the desired homotopy $H' : D^m \times [0, 1] \rightarrow Y$ to be

$$H'(x, t) = h((1 + t)x)$$

It is left to check that this actually defines a continuous extension of the homotopy H and that it starts with f .

-> wenn zu kurz, dann hier ausführlicher □

Since the definition of CW-complexes in Mathlib uses n -dimensional cubes instead of closed balls as preimage of a closed cell under the characteristic map, we will state the *HEP* for cubes relative to their boundary.

Corollary 2.16. $\forall n \in \mathbb{N}$: $([-1, 1]^n, \partial[-1, 1]^n)$ has the HEP.

Proof. This statement follows from the observation, that D^m is homeomorphic to $[-1, 1]^n$. One constructs a homeomorphism via the radial projection, which maps the boundary of the sphere to the boundary of the cube. All points in the interior are scaled along the rays. In that way, the cube inherits the *HEP* from the sphere. □

In the next step, we will prove, that for every relative CW complex (X, A) with skeleton X_n and for all $n \in \mathbb{N}$, the pair (X_{n+1}, X_n) has the *HEP*. – The key-idea of the proof is to apply the retraction criterion. To construct the retraction, we apply a retraction for each of the added cells relative their boundary at the same time. On the base space, we use the identity. Later, we will need this result in order to construct a retraction on the entire CW-complex.

Theorem 2.17. *Let (X, C) be a relative CW-complex with skeleton X_n , then $\forall n \in \mathbb{N} : (X_{n+1}, X_n)$ has the *HEP*.*

Before provig this theorem, we need some preparation. The key-idea of the proof is, to apply the retraction criterion, i.e. we want to construct a map on $X_{n+1} \times [0, 1]$.

Proof. We proof the *HEP*, using the retraction criterion.

First, we define the map and then proof afterwards, that it has the required properties, i.e. is surjective and the weak topology agrees with the inuced topology of the quotientmap. For the definition of the map, we use the retraction r_C from the *HEP* for cubes relative boundary, which is defined on $[-1, 1]^n \times [0, 1]$. Even tough r_C depends on the dimension of the cube, for reasons of convenience, we use the same notation for all of those maps. Let J_n denote the indexing set for cells in dimension $n \in \mathbb{N}$ and $Q_i^n : D^n \rightarrow X$ be the charakteristic map of the i th n -cell.

Informally spoken, the map r

$$r : X_{n+1} \times [0, 1] \rightarrow X_{n+1} \times \{0\} \cup X_n \times [0, 1]$$

$$(x, t) \mapsto \begin{cases} r_C(), & \text{if } t = 0, \\ H(x, t), & \text{else} \end{cases}$$

□

3 Lean and mathlib

After taking a closer look at the informal mathematics behind my formalisation in the previous chapter, I now want to give a brief overview of what Lean is. I will explain some of the conventions and concepts used in formalization that are relevant to this project.

Lean is a proof assistant, i.e. a tool that checks and confirms the individual reasoning steps of mathematical proofs written in Lean’s formal language. It verifies that assumptions, axioms, functions, and theorems are only applied where permitted, and that new results are derived using valid logical inferences. In *The Lean Theorem Prover (system description)* the main designers of Lean explain that the correct verification of mathematical proofs in Lean rests upon a “small trusted kernel based on dependent type theory” (leanTP). I will elaborate the term “dependent type theory” later in this section.

Lean is one of several proof assistants, alongside others such as Rocq or Isabelle. However, as Alex Kontorovich elaborates in *The Shape of Math to Come* [Kon25], it is remarkable how quickly Lean is gaining relevance, not only in computer science and mathematical logic, but also in mainstream mathematical research.

Since July 2023, the non-profit *Lean Focused Research Organization (FRO)* has supported and coordinated the ongoing development of the language. Its stated goal is “enhanced scalability, usability, documentation, and proof automation while guiding Lean toward long-term self-sustainability” [FRO]. The same source also provides a brief overview of Lean’s development history. The first version was created by *Leonardo de Moura* in 2013. Since then, the language itself has evolved. The first stable release of Lean 4 appeared in September 2023, and both the community and the digital library *mathlib* have grown considerably. Today, mathlib 4 contains more than 2,000,000 lines of code and has over 770 contributors. Current statistics can be found at [1].

3.1 Lean’s Dependent Type Theory

The digital book *Theorem Proving in Lean 4* teaches how to verify proofs in Lean and explains the underlying logical system. Its second chapter, on *Dependent Type Theory*, serves as the basis for the following description. The first and probably most important fact is that every expression in Lean has a type. Using the command `#check`, one can display the type of an expression. In the example below, I first define `n` to be the natural number 3 and then check its type, which is `ℕ` by the definition above. As stated, every expression in Lean has a type, so the natural numbers themselves can be checked as well. Their type is called `Type`. The classical objects $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, as well as the finite-dimensional Euclidean spaces, all have type `Type`. Continuing the same procedure and checking the type of `Type`

yields `Type : Type 1`. This can be repeated indefinitely: in general, `Type k` has type `Type (k+1)` for every natural number k . I want to present one more Type, which is called `Prop`. As the name already implies, `Prop` is the type used for propositions, e.g. $2 = 1$. We can combine propositions to build new ones. For example we can connect two propositions with a logical "and" in order to obtain a new expression of type `Prop`. `Prop` itself has type `Type`.

```
def n : ℕ := 3
#check n
n : ℕ
#check ℕ
ℕ : Type
#check EuclideanSpace ℝ (Fin 3)
EuclideanSpace ℝ (Fin 3) : Type
#check Prop
Prop : Type
#check Type
Type : Type 1
#check 2 = 1
2 = 1 : Prop
#check Prop
Prop : Type
```

Type theory not only allows one to define an element, such as the natural number 3 above, but also to construct new types from existing ones. Functions are one such example. To illustrate this, I start by choosing two arbitrary but fixed types α and β . Writing `Type*` admits types of any universe. Lean then automatically introduces a so-called universe variable, here `u_1`. A function with domain α and codomain β has type $\alpha \rightarrow \beta$. This newly constructed type again has a type of its own, whose universe is the maximum of the universe levels of α and β .

```
variable (α β : Type*) (f : α → β)
#check α
α : Type u_1
#check f
f : α → β
#check α → β
α → β : Type (max u_1 u_2)
```

Functions such as the addition of natural numbers, which one would usually write as a map $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, have in Lean the type $\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}$. This means that after applying a single natural number to the addition function, we obtain a function $\mathbb{N} \rightarrow \mathbb{N}$ that adds the first number. Since $\mathbb{N}$ has type `Type`, the reasoning of the previous paragraph shows that the function type $\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}$ again has type `Type`. Note the distinction between the two levels: the addition function itself has type $\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}$, whereas this function type in turn has type `Type`.

```
#check Nat.add
      Nat.add :  ℕ → ℕ → ℕ
#check ℕ → ℕ → ℕ
      ℕ → ℕ → ℕ :  Type
```

Even though I do not want to illustrate in detail, what makes the type theory of lean a dependent type theory, let me

3.2 Structures

3.3 Junk Values

If you write a division by zero in a formalized theorem, Lean returns the result 0.

```
example : 1 / 0 = 0 := by norm_num
```

This makes division in Lean always well-defined, even though certain inputs have no genuine mathematical meaning. As explained in [MLS25], the use of so-called *junk values* is common practice. The same convention applies, for instance, to the inversion of square matrices, which is always defined and returns the zero matrix whenever the matrix is singular, or to the derivative of a non-differentiable function, which is defined to be zero. Many further examples of this kind exist. At first glance this may seem concerning, since such values have no mathematical justification and could appear to enable incorrect proofs. Indeed, one can state and prove propositions that, in classical mathematics, would look false or ill-defined. This issue is resolved by adding suitable hypotheses such as $n \neq 0$ or `Differentiable k f` to any statement about the object in question. They ensure we only use the result of the theorem in applicable situations, such as, if our number is nonzero or our function is differentiable. Consequently, although the convention may seem alarming initially, relying on junk values is standard practice in Lean formalizations, precisely because it simplifies both definitions and statements. Returning to the first example, it is what allows division to be expressed as a total map $\mathbb{Q} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}$.

3.4 API lemma

4 Lean formalisation of the Homotopy Extension Property

4.1 CW-complexes in Mathlib

1. NOTIZEN:
2. Zellen im CW complex sind würfel anstatt spheres
3. CW -complexe leben in einem größeren Raum ->

4.2 HEP Definition

Already in the mathematical section, the definition of the *homotopy extension property* was quite long and technical. The same is true for its formalization. Currently I have three versions of the definition in my project, since we decided within the work progress, that different phrasings might be more applicable. I will later on discuss the pros and cons for each of the version. The first definition is the following, called HEP:

```
def agreeOn.{u} {X Y : Type u} (f : X → Y) (H : X × ℝ → Y) (A : Set X) : Prop :=
  ∀ (a : X), a ∈ A → f a = H (a, 0)

structure HomotopyExtension.{u} {X Y : Type u} [TopologicalSpace X]
  [TopologicalSpace Y] (H' : X × ℝ → Y) (f : X → Y)
  (H : X × ℝ → Y) (A : Set X) (rangeH' : Set Y) : Prop where
ContinuousOn : ContinuousOn H' { p : X × ℝ | p.2 ∈ unitInterval }
range : { p : X × ℝ | p.2 ∈ unitInterval }.MapsTo H' rangeH'
agreef : (∀ (x : X), f x = H' (x, 0))
agreeH : (∀ (a : A) (t : unitInterval), H (a,t) = H' (a, t))

def HEP.{u} (X : Type u) [TopologicalSpace X] (A : Set X) : Prop :=
  ∀ (Y : Type u) [TopologicalSpace Y] (rangeH' : Set Y), ∀ (f : X → Y),
  ∀ (H : X × ℝ → Y),
  Continuous f → Set.range f ⊆ rangeH' →
  ContinuousOn H {p : X × ℝ | p.1 ∈ A ∧ p.2 ∈ unitInterval} →
  {p : X × ℝ | p.1 ∈ A ∧ p.2 ∈ unitInterval}.MapsTo H rangeH' →
  agreeOn f H A
  → ∃ H' : X × ℝ → Y, HomotopyExtension H' f H A rangeH'
```

To shorten the definition a little and in order to make it more readable, I first defined `agreeOn`. This tells, that the function `f` and the homotopy `H`, which we want to extend, have the same values on the subset `A` respectively $A \times 0$. So it ensures, that the two maps are compatible and it makes sense to ask for an extension. Secondly I constructed the structure called `HomotopyExtension`, which combines all the properties, our desired homotopy should have. Namely, continuity, that its image lies in the right space and

that it agrees with the two functions given in the begin. Making this a structure, instead of a simple definition has the advantage, that it comes with the projection maps for each property.

If we now take a closer look at the definition of the homotopy extension property, it is pretty similar to the mathematical definition theorem 2.1 of the *HEP*. Still, there is one point, I want to discuss. Defining the homotopies as a map on $X \times \mathbb{R}$ instead of $X \times \text{unitInterval}$ or $A \times \text{unitInterval}$, makes it easier to work with it. For instance consider the definition `agreeOn`. Assume, the homotopy H was defined as a map on $A \times \text{unitInterval}$, then one had to provide a proof of $a \in A$, every time one wants to apply a point $(a : X)$ to the function. Also in the second argument, we can apply any real number to it, without the need to provide a prove that this real number lies between zero and one.

Still, this does not change anything about the defined object. `ContinuousOn H' { p : X × ℝ | p.2 ∈ unitInterval }` and $\forall (a : A) (t : \text{unitInterval}), H(a, t) = H'(a, t)$ make it irrelevant, what happens outside of the compact interval, but instead put precise focus on what we need for our definition.

Moreover, the set `rangeH'` has no equivalent in the informal definition, but is a result of formalisation. `rangeH'` is a set of the codomain-type Y and as the name indicates, the image of the extended homotopy H' is contained in it. Together with adding this set to the definition, there are two hypotheses, which we need to add. H' cannot have image in `rangeH'`, if not f and H already have image in there.

The win of adding `rangeH'` to the definition is again, that we can avoid working with subtypes. If we want to construct an extended homotopy, which has image in a subtype, we now get a map between types together with the result, that the image lies in the required subset. Nevertheless, in most cases, one wants to choose `rangeH'` to be all of Y . In Lean this can be expressed as `@ Set.univ Y`. The two additional assumptions are then statements, which can be easily proven by `tauto`. Since this extra work is very manageable, I prefer this definition of *HEP* over the following version called *HEPY*, where `rangeH'` is set to be the entire codomain Y . Even though this one is shorter and closer to the informal version. The `leamm HEP_iff_HEPY` proofs, that those two definitions are equivalent

```
structure HomotopyExtensionY.{u} {X Y : Type u} [TopologicalSpace X]
  [TopologicalSpace Y] (H' : X × ℝ → Y) (f : X → Y)
  (H : X × ℝ → Y) (A : Set X) : Prop where
  ContinuousOn : ContinuousOn H' { p : X × ℝ | p.2 ∈ unitInterval }
  agreef : ∀ (x : X), f x = H' (x, 0)
  agreeH : ∀ (a : A) (t : unitInterval), H (a,t) = H' (a, t)
```

```
def HEPY (X : Type u) [TopologicalSpace X] (A : Set X) : Prop :=
  ∀ (Y : Type u) [TopologicalSpace Y], ∀ (f : X → Y), ∀ (H : X × ℝ
```

```

→ Y), Continuous f →
ContinuousOn H {p : X × ℝ | (p.1 ∈ A) ∧ (p.2 ∈ (unitInterval))}
→ agreeOn f H A
→ ∃ (H' : X × ℝ → Y), HomotopyExtensionY H' f H A

```

The last aspect, I want to elucidate, before progressing to the final third version of the definition is, that I fixed the Type of X and Y to be the same, namely both are `Type u`, for some arbitrary universe u . This was needed, in order to – `ToDo`

Let us now compare this definition with the later version. HEP' :

```

open Set.Notation
def HEP' {Y : Type*} [TopologicalSpace Y] (X A : Set Y) : Prop :=
  HEP X (X ↓∩ A)

```

The notation $(X \downarrow \cap A)$ means, consider $(X \cap A)$ as a set of Type `Set X`. Then HEP' , is defined in terms of HEP for the pair X and $(X \downarrow \cap A)$. The difference to the previous versions of the definition is, that in the definition HEP , A is a subset of the type X . That means that the pair of spaces does not live in an ambient space. Whereelse, in the definition HEP' , both sets X and A are sets of the same type Y .

We decided to go for the second version of the definition, when we wanted to phrase the *homotopy extension property* for pairs of consecutive skeletons. A skeleton of a relative CW complex in Lean is a subcomplex, whose carrier is a subset of an ambient space. Also it turned out to be the nicer way, to prove lemmas about closed balls and spheres, because I do not have to consider the sphere as a subset of "Type closedBall" anymore, but just as the points with distance one to the origin in a metric space.

Note, that in this definition the functions f and H , which we want to extend are defined on the subtype $\uparrow X$, respectively $\uparrow X \times \mathbb{R}$. Working with those subtypes caused some extra work in the proof of `retraction_criterion_closed'` and also `HEP_trans'`. One could avoid this by modifying the definition HEP' in a way, that it is not defined using HEP , but constructed explicitly with functions defined on Y , respectively $Y \times \mathbb{R}$. The definition in that case would be longer, since for example continuity of f would change to a `ContinuousOn`-statement. Also, it would be more work to prove equivalence between HEP and this newer version, that it was now, where we basically got it for free. Nevertheless I am convinced, it would improve the formalisation of the definition.

To sum up, I value the primed version HEP' to be the most useful one to work with. Optimally with the discussed change, to write the definition explicitly with functions defined on the entire space instead of subspaces. Still, I don't think this outdoes the definition HEP , in all accounts. Since the statements become longer and also the proof of the retraction criterion would have been longer and more tedious.

4.3 Retraction Criterion

Before elucidating the retraction criterion its formalisation, we look at the definition of a retraction. The classical way to define a retraction r from B to A is, as in theorem 2.7, via precomposition with the inclusion $A \hookrightarrow B$. As already in the formalisation of the definition *HEP*, we again want to avoid working with subtypes. So instead of defining a map from B to A , for the formalisation, the retraction map is defined from an ambient type X to itself. The structure `RetractionOn` bundles all properties, which make this map a retraction.

```
structure RetractionOn.{u_1} {X : Type u_1} [TopologicalSpace X] (r :
  X → X) (B A : Set X) : Prop where
  subset : A ⊆ B
  continuousOn : ContinuousOn r B
  mapsTo : B.MapsTo r A
  fixesOn : ∀ a ∈ A, r a = a
```

As mentioned before, that the map r is a map from a Type X into itself and it is not a retraction on all of its domain. As the name `RetractionOn` implies, it just "retracts" a subset B to a subset A of the Type X . The first property `subset` states, that A is a subset of B . Note, that leaving out this property in our structure would define a more general class of selfmaps, which does not necessarily satisfy $r(B) = A$ anymore. The property `continuousOn` ensures, it is a continuous Map on the relevant subset B . The codomain of the retraction is supposed to be A , so the property `mapsTo` claims, that r maps all elements of B to an element in A . Furthermore the classical definition states, that precomposition with the inclusion $A \hookrightarrow B$ should give the identity id_A . In our case there is not really an inclusion map, since both - image and preimage of r - are the same Type X . So in our setting this condition translates to, that each element in A needs to be fixed by applying r . This property is called `fixedOn`.

Recall, that the retraction criterion was a corollary of two lemmas. The same structure can be found in the formalisation. theorem 2.5 has two implications, one of which requires closedness. In order to only assume closedness when needed, in my project each implication has its own lemma, called `if_HEP_then_extension` and `if_extension_then_HEP`. The formalised proofs are quite unspectacular and just follow the pattern of the informal proof found in the above section/chapter?. The same is true for the second lemma theorem 2.6 as well as the retraction criterion theorem 2.8 itself.

```
lemma retraction_criterion_closed.{u} {X : Type u} [TopologicalSpace
  X] {A : Set X} (hA1 : IsClosed A) :
  HEP X A ↔ ∃ r : X × ℝ → X × ℝ, RetractionOn r {p | p.2 ∈
    unitInterval} {p | p.2 = 0 ∨ p.1 ∈ A ∧ p.2 ∈ unitInterval}
```

The retraction in this above lemma is a map on the product $X \times \mathbb{R}$, where X is the bigger space of the space pair from the homotopy extension property. Even though this lemma is phrased for HEP, we can also apply it to HEP':

```
variable {Y : Type u} [TopologicalSpace Y] (X A : Set Y) (h : HEP' X
  A)
#check if_HEP_then_retraction h
```

Even though this yields a retraction for the relevant spaces, this does not quite assemble the idea of HEP'. Instead I formalised the lemma `retraction_criterion_closed'`, where the retraction is a map from the ambient space $\times \mathbb{R}$ to itself. This is realised in the following lemma. In the lean file you can again find the forward implication without closedness assumption.

```
lemma retraction_criterion_closed'.{u} {Y : Type u} [
  TopologicalSpace Y] (A X : Set Y) (hAX : A  $\subseteq$  X) (hA1 : IsClosed
  A) :
  HEP' X A  $\leftrightarrow$   $\exists$  s : Y  $\times$   $\mathbb{R}$   $\rightarrow$  Y  $\times$   $\mathbb{R}$ , RetractionOn s {p | p.1  $\in$  X  $\wedge$ 
  p.2  $\in$  unitInterval} {p | p.1  $\in$  X  $\wedge$  p.2 = 0  $\vee$  p.1  $\in$  A  $\wedge$  p.2  $\in$ 
  unitInterval}
```

I proved the `retraction_criterion_closed'` using the `retraction_criterion_closed`. Explicitly, for the forward direction, I obtained a retraction r as in `ToDo`. In order to extend this map r to a map $s : Y \times \mathbb{R} \rightarrow Y \times \mathbb{R}$, I precomposed in the first component with a map called `mapYX` : $Y \rightarrow X$, which is the identity on the set X and maps all other points of Y to an arbitrary junk value in the `unitInterval`. `Subtype.val` sends $(r\ x).1 \in X$, to the corresponding point in Y . Afterwards, I had to confirm this new defined map r indeed is a retraction, i.e. it is continuous on the relevant set, maps to the required one and the necessary values remain unchanged under applying r .

For the backward direction I start with a retraction $s : Y \times \mathbb{R} \rightarrow Y \times \mathbb{R}$ and want to construct $r : \uparrow X \times \mathbb{R} \rightarrow \uparrow X \times \mathbb{R}$, so applying `retractioncriterionclosed` yields, the claim. One might expect this direction to be easier, since we don't have to construct any additional values, but just restrict s , to the desired space. Sadly these dreams shatter at the point, where we want the image to lie in $X \times I \subset Y \times \mathbb{R}$. s is a retraction on $\cdot$, so the `RetractionOn.mapsTo` property gives no information about, where in $Y \times \mathbb{R}$ we are, if the second coordinate comes from outside the unit interval. I solved this issue by precomposing the real coordinate with the map $c : \mathbb{R} \rightarrow \mathbb{R} := \text{fun } t \mapsto (\text{Set.projIcc } (0 : \mathbb{R}) 1 \text{ zero_le_one } t)$. `Set.projIcc`, is a continuous map from `mathlib`, which retracts a linear ordered Type to a closed interval in it. This precomposition makes, that for every point, which we apply to s , the `mapsTo` property of s is applicable. This extra work, is a downside of working with types instead of subtypes. It would have been avoided, if I defined a homotopy only on $Y \times \text{unitInterval}$, instead of $Y \times \mathbb{R}$.

I want to point out, that even though the two versions of the retraction

criterion are phrased with two different versions of the definition, this was more a choice based to style, than on technical need. Also, I experienced both of these versions to be valuable. While it might be advisable to just choose the definition `HEP'`, I see usage in both versions of the retraction criterion. The usage of `retractioncriterionclosed` is underlined by the the proof, that the pair of two consecutive skeletons satisfies the *HEP*. I constructed a retraction $r: (\text{SkeletonLT } X \text{ } m) \times \mathbb{R} \rightarrow (\text{SkeletonLT } X \text{ } m) \times \mathbb{R}$. The key theorem for the construction of this continuous map was the lemma `Topology.IsQuotientMap.continuous_lift_prod_left`, which is a formalisation of Whitehead's Theorem about products of quotient maps with a locally compact factor. If I worked with ambient spaces, I would have lost the surjectivity of the quotient map and would not have been able to apply this theorem, the way it can be found in `mathlib`.

4.4 Preservation of the HEP

Besides the retraction criterion I formalised two more theorems, which are useful to prove, that a pair of topological spaces has the *HEP*. One is, that the *HEP* is preserved under homeomorphisms. The other is sort of a transitivity property, i.e. if $(A1, A2)$ and $(A2, A3)$ are pairs of topological spaces and both of them satisfy the *HEP*, then also the pair $(A1, A3)$ has the *HEP*.

```
lemma HEP_trans.{u_1} {X : Type u_1} [TopologicalSpace X] (A1 A2 A3 :
  Set X) (h21_sub : A2 ⊆ A1) (h12_hep : HEP' A1 A2) (h32_sub : A3
  ⊆ A2) (h23_hep : HEP' A2 A3) : HEP' A1 A3
```

For the formalisation of the proof, I had the choice between two different attempts. Instead of applying the retraction criterion to all the *HEP* statements, and then using the two retractions from my assumptions to construct the desired retraction, I proved the lemma by checking the definition. Another positive side effect of this choice was, that I don't have to assume closedness of any Sets, which would be required to apply the retraction criterion.

4.5 HEP for discs and cubes

The goal of the file `HEP_ball_cube.lean` is, to prove the homotopy extension property for balls and cubes relative to their boundary. These are not just the first interesting space pairs, we want to prove the *HEP* for, but also the first step in the prove of the *homotopy extension property* for relative CW complexes.

We start with the balls: As in the prove of theorem 2.15, I first constructed the auxiliary function `auxfun` combining the information of the given maps f and H . Then this function is used to explicitly define a desired homotopy H' , which we need to check the definition for *HEP*.

```

variable {m : ℕ} {X Y : Type} [TopologicalSpace X] (C : Set Y)
(f : (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) → Y)
(H : (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) × ℝ →
Y)

def aux_fun : EuclideanSpace ℝ (Fin m) → Y := fun p =>
  if hp : norm p ≤ 1 then f ⟨p, by simp[hp]⟩
  else H (⟨((norm p)-1 : ℝ) · p, inv_norm_smul_mem_unitClosedBall p⟩,
    norm p - 1)

def H' : (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) × ℝ
→ Y :=
  fun p => (aux_fun f H) ((1 + p.2) · p.1)

```

I split up the `ContinuousOn` proof of `H'` in three steps. I started with a lemma, that states `aux_fun` is continuous on the ring $\{x \mid 1 \leq \|x\| \leq 2\} \subset \mathbb{R}^m$ and used this to proof continuity of `aux_fun` on the ball with radius two. The ball with radius two can be written as the union of a ball with radius one and the ring. Continuity on both closed subsets yields continuity on their union. The continuity of `H'` follows from composing two `ContinuousOn` functions. The proofs for all desired properties of `H'` are not hard. Solving, rewirting and confirming inequalities make up big parts of them. All those results combined proof the lemma `HEP_disc_boundary`. The primed version `HEP_disc_boundary'`, is definitionally equal.

```

lemma HEP_disc_boundary : ∀ (m : ℕ),
  HEP (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) (Metric.
    sphere ⟨0, by simp⟩ 1) := by ...

lemma HEP_disc_boundary' : ∀ (m : ℕ),
  HEP' (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) (Metric.
    sphere 0 1) := HEP_disc_boundary

```

The `Metric.sphere` in Lean has type `Set α`, where α is the Type of the center (in our case the 0 in an n -dimensional euclidian space). By the way `HEP` is defined, in `HEP_disc_boundary` the sphere is a subset of the closed ball and we have to add the `by simp` proof, to construct the element of the subtype. On the other hand, for `HEP'` both spaces live in the same ambient space. We do not have to add a proof Lean interpretes the zero as the origin of $\mathbb{R}^m$. Here we see an advantage of the `HEP'` definition in action, since we can reduce the use of subtypes.

In the formalisation of (relative) CW-complexes, each cell is the image of a closed ball in $(\text{Fin } m \rightarrow \mathbb{R})$ for some m under a characteristic map. Same as `EuclideanSpace ℝ (Fin m)`, also `Fin m → ℝ` is a way to formalize a m -dimensional real vectorspace. The relevant difference between them is, that the `EuclideanSpace` comes with the Euclidian norm, wherelse on the other version, the default norm is the maximum norm. So the closed ball, which is a preimage of a closed cell in a CW complex, is something we usually would call a cube, not

a ball. To prove the HEP for those cubes, we want to apply the fact, that a m -dimensional ball is homeomorphic to a cube of the same dimension. Instead of explicitly construction this homeomorphism, I wanted to obtain it from the following mathlib lemma. It states, that "if s is a convex bounded set with a nonempty interior in a real normed space, then there is a homeomorphism of the ambient space to itself that sends the interior of s to the unit open ball and the closure of s to the unit closed ball."

```
exists_homeomorph_image_interior_closure_frontier_eq_unitBall.{u_1}
  {E : Type u_1} [NormedAddCommGroup E] [NormedSpace ℝ E] {s : Set
    E} (hc : Convex ℝ s) (hne : (interior s).Nonempty) (hb :
    Bornology.IsBounded s) :
  ∃ h, ↑h '' interior s = ball 0 1 ∧ ↑h '' closure s = closedBall 0
    1 ∧ ↑h '' frontier s = sphere 0 1
```

It is quite obvious, that the `Metric.closedBall`, for which we proved the HEP already, is closed, convex and nonempty and therefor satisfies all assumptions. Still we don't want to apply the above lemma directly on this. The statement would be trivial and we just obtained the identity map on the m -dimensional Euclidian space. Instead we first map this closed ball to $\text{Fin } m \rightarrow \mathbb{R}$ via the homeomorphism

```
def homeo_euclid_max : EuclideanSpace ℝ (Fin m)  ≃t (Fin m → ℝ)
  where
  toFun := fun p ↦ p
  ...
```

`homeo_euclid_max` maps each point to itself while changing the norm of the space. So the image of the `Metric.closedBall` is still the same round, convex set, even though it is not the same as a closed ball in the codomain-space. Applying the lemma to the image of the `Metric.closedBall` yields a homeomorphism from the $\text{Fin } m \rightarrow \mathbb{R}$ to itself:

```
def G : (Fin m → ℝ) ≃t (Fin m → ℝ) := (
  exists_homeomorph_image_interior_closure_frontier_eq_unitBall
  convex_euclid_to_max_ball nonempty_euclid_to_max_ball
  bounded_euclid_to_max_ball).choose
```

The composition of those two homeomorphisms is especially a partial homeomorphism `EuclideanSpace ℝ (Fin m)` to $\text{Fin } m \rightarrow \mathbb{R}$ with source the closed ball and target the closed cube. Now applying the lemma `PartialHomeomorph_HEP'` proves

```
lemma HEP_cube_boundary' (m : ℕ) : HEP' (closedBall (0 : (Fin m →
  ℝ)) 1) (sphere 0 1) := by ...
```

```
lemma HEP_cube_boundary (m : ℕ) : HEP (closedBall (0 : (Fin m → ℝ))
  1) (sphere ⟨0, by simp⟩ 1) := by
  exact HEP_cube_boundary' m
```

Before concluding this section, I want to point out, that here the definition of `HEP'` matches very good the structure of the partial homeomorphis. When I first proved this result - before defining `HEP'` - for `HEP`, I had to show that the `Subtze.val` of a sphere in the closed ball is the same as the sphere in the ambient space and the same for the closed ball itself. With `HEP'` we avoid this subtype coercions and the proof is more straight forward. From this result the statement for `HEP` follows directly.

4.6 Skeleton is a quotient

(Relative) CW complexes are formalised as a subset of an ambient space. The topology is given by explicitly saying, a closed set needs to be closed, when intersected with the base space or any closed cell it is closed in those spaces. On a skeleton or any finite dimensional CW complex is the same topology, which we also obtain from gluing together discs along their boundary and then considering the quotient topology on the complex.

From the previous file I already have retractions defined on closed balls times the unit interval. Together with the fact, that I can "unglue" my CW complex into a basespace and closed balls in all dimensions, I can construct the retraction on a skeleton.

The formalisation of this step builds on the following mathlib lemma. It replaces the theorem, which states, that the product of a quotient map with the identity on a locally compact space, is again a quotient map.

```
theorem Topology.IsQuotientMap.continuous_lift_prod_left {X₀ : Type
  u_1} {X : Type u_2} {Y : Type u_3} {Z : Type u_4} [
  TopologicalSpace X₀] [TopologicalSpace X] [TopologicalSpace Y] [
  TopologicalSpace Z] [LocallyCompactSpace Y] {f : X₀ → X} (hf :
  IsQuotientMap f) {g : X × Y → Z} (hg : Continuous fun (p : X₀ ×
  Y) => g (f p.1, p.2)) :
  Continuous g
```


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